

Current Position

02/2022 – present | **Ph.D. • Formal AI Safety**
📍 Technical University of Munich, Germany
with “*summa cum laude*”



Education

09/2025 – 02/2026 | **Visiting Researcher**
📍 University of California, Irvine, USA



10/2019 – 12/2021 | **Master of Science**
📍 Technical University of Munich, Germany
Computer Science • with high distinction



Winter 2020 | **Erasmus Exchange Semester**
📍 University of Sussex, United Kingdom
Computer Science



10/2016 – 08/2019 | **Bachelor of Science**
📍 Technical University of Munich, Germany
Computer Science • with merit



Selected Publications

- [1] **On Convex and Non-Convex Abstraction Refinement Techniques Guaranteeing Safety of Artificial Intelligence.** Tobias Ladner. *Ph.D. Thesis. Technical University of Munich.* 2026.
- [2] **Provably Explaining Neural Additive Models.** Shahaf Bassan*, Yizhak Yisrael Elboher*, Tobias Ladner*, et al.. *ICLR.* 2026.
- [3] **Out of the Shadows: Exploring a Latent Space for Neural Network Verification.** Lukas Koller, Tobias Ladner, and Matthias Althoff. *ICLR.* 2026.
- [4] **Formal Verification of Graph Neural Networks with Uncertain Node Features and Uncertain Graph Structure.** Tobias Ladner, Michael Eichelbeck, and Matthias Althoff. *TMLR.* 2025.
- [5] **Explaining, Fast and Slow: Abstraction and Refinement of Provable Explanations.** Shahaf Bassan*, Yizhak Yisrael Elboher*, Tobias Ladner*, Matthias Althoff, and Guy Katz. *ICML.* 2025.
- [6] **Fully Automatic Neural Network Reduction for Formal Verification.** Tobias Ladner and Matthias Althoff. *TMLR.* 2025.

Professional Contributions and Activities

- Administrator of toolbox [CORA](#)
- Repeatability co-chair of [ARCH-COMP](#) (2023-2026) and [VNN-COMP](#) (2025-2026)
- Invited talks at [Munich Datageeks](#), [University of Aalborg](#), and at [LEISYS](#): “Formal Verification of Neural Networks in Safety-Critical Environments”
- Hobby projects: [Android Reddit Wallpaper](#), [Covid Data Visualization](#)

Further Work Experiences

Various internships as a software engineer: [INVENOX](#) (2017), [ARZ](#) (2013-2016), and [AKH](#) (2012). Tutor at [Technical University of Munich](#) (2018, 2019). Civil service at [Austrian Red Cross](#) (2015-2016). Ski instructor (holidays 2010-2020).

About Me

Tobias Ladner
📍 Munich, Germany

Specialization

AI Safety,
Formal Methods,
Adversarial Robustness,
Set-Based Computing,
Automated Reasoning,
Interpretability

Programming

python, matlab, lean,
latex, java, kotlin, ocaml,
html, css, js;
agentic programming

Languages

German (native),
English (C2)

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🌐 [Website](#)

🌐 [LinkedIn](#)

🎓 [Google Scholar](#)

🐙 [GitHub](#)